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ORIGINAL PAPER

Reduction in EHL Friction by a DLC Coating

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Abstract This work evaluates the influence of DLC coatings on the frictional behaviour of lubricated line contacts in the fluid friction regime. Experiments on a twin-disc machine showed a significant reduction in friction using DLC coated discs instead of uncoated ones. The experimental results are compared by calculations using a three-dimensional thermal elastohydrodynamic simulation model. According to the calculations, the lower heat conductivity of DLC coatings is the main reason for the reduction in friction. This is because of the higher temperature level in the lubricating gap, which results in a reduction in the effective lubricant viscosity and consequently in reduced frictional shear stress. Lower bulk temperatures of the discs are a further consequence of the low heat conductivity of DLC coatings. The resulting lower temperature at the inlet zone of the contact in the steady-state conditions also leads to larger lubricating gaps and to a further reduction in friction.

Keywords Diamond-like carbon (DLC) · Twin-disc machine · Thermal elastohydrodynamics · Non-Newtonian flow · Line contact · Friction reduction

List of symbols

c	Specific heat capacity [J/(kg K)]
E_{red}	Reduced Young's modulus (N/m^2)
f	Coefficient of friction (−)
F_{f}	Friction force (N)
F_{n}	Normal force (N)
h	Film thickness (m)
h_0	Nominal film thickness (m)
h_{cr}	Critical film thickness (m)
h_{def}	Deformed gap height (m)
h_{min}	Minimum film thickness (m)
$h_{\delta w}$	Local deformed gap height (m)
k	Coefficient of thermal partition
n	Rotation speed (1/min)
p	Pressure (N/m^2)
$p_{\mathrm{c,lim}}$	Plastic flow pressure (N/m^2)
$\dot{q}$	Heat flux density (W/m^2)
R	Radius (m)
t	Time (s)
u, v	Velocities in x and y direction (m/s)
x, y, z	Cartesian coordinates (m)
α_{p}	Pressure viscosity coefficient (m^2/N)
β_{th}	Coefficient of volume expansion (1/K)
$\dot{\gamma}$	Shear rate (1/s)
η	Dynamic viscosity (Pas)
η^*	Effective dynamic viscosity (Pas)
θ	Gap fill factor (−)
ϑ	Temperature (K, °C)
λ	Thermal conductivity [W/(m K)]
ρ	Density (kg/m^3)
τ	Shear stress (N/m^2)
τ_{c}	Eyring shear stress (N/m^2)
τ_{f}	Friction shear stress (N/m^2)
Φ^{p}	Pressure flow factor (−)
Φ^{s}	Shear flow factor (m)
Ω	Calculation domain (m^2)

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Frequently used indices

c	Contact
cav	Cavitation
h	Hydrodynamics
liq	Liquid
mix	Mixed

1 Introduction

Under the term of diamond-like carbon (DLC) coatings, amorphous carbon coatings are referred. According to VDI-code 2840 [1], DLC coatings are classified in hydrogen-free (a-C) and hydrogen containing (a-C:H) coatings with or without metal doping (a-C:Me and a-C:H:Me). Coatings with predominating sp^3 hybridised carbon are called tetrahedral amorphous carbon coatings (ta-C and ta-C:H). Depending on the requirement of the coating, additional elements such as silicon, oxygen or chrome can be introduced. Generally, DLC coatings feature high hardness and a good stability against thermal and chemical stress.

DLC coatings are often used in lubricated tribological contacts in order to enhance the strain, if a heavy mixed friction regime is expected. Typical applications for DLC coatings are fuel-lubricated contacts in fuel injection pumps [2], gear pairs [3] or the improvement of the start-up or run-out performance of sliding bearings [4]. Besides the enhancement of strain, the effect of DLC coatings on the frictional behaviour in lubricated contacts was the subject of investigation. Evans et al. [5] presented Stribeck curves based on experiments with a ball on plate tribometer for different types of coating. They used two DLC coatings containing hydrogen (WC/a-C:H). Poly-alpha-olefin (PAO) was used as lubricant. In the regime of boundary friction, the lowest friction level was observed with DLC coated discs. Yang et al. [6] used an oscillating pin on disc tribometer to investigate the influence of DLC coated discs (a-C:H/W) on the friction performance for dominating boundary friction. They used PAO with and without additives influencing the friction and wear behaviour. Experiments using non-additivated lubricants showed a clear reduction in friction for the use of DLC coated discs, such as mentioned in [5]. The experiments carried out with the additivated lubricant did not show such unambiguous results. The friction coefficients measured for uncoated discs were reduced by the use of the glycerol-mono-oleate (GMO) as friction modifier. In contrast, the friction of DLC coated discs was enhanced. The use of MoS_2 as additive showed a reduction in the friction level for all experiments. However, the change was quite higher for uncoated discs. Therefore, no clear conclusion according to the reduction in friction for the DLC coatings was possible. A positive effect of MoS_2 additives on the frictional behaviour of DLC (a-C:H)-coated samples is reported in Kogovsek et al. [7] for experiments with a ball on disc tribometer. The balls, as well as the discs, were DLC coated for the experiments. Comparing experiments with uncoated samples was not performed. The presented studies show the possibility of friction reduction by use of DLC coatings. However, a friction reduction is not mandatory. The chemical reactions between the DLC coating and the lubricant or rather the contained additives are important factors.

Regarding DLC coatings, it has also been found that the frictional behaviour is not only influenced in the mixed friction regime, but also in dominating fluid friction. Grossl [8] and Evans et al. [5] observed that the friction of DLC coated samples is smaller compared to uncoated samples, even beyond the mixed friction regime. The experiments in [5] were performed without boundary friction ratios for which reason other mechanisms have to be responsible for friction reduction. For the authors [5], the different wetting behaviour of the lubricant for uncoated and DLC coated surfaces was supposed to be the reason for low friction. Thus, the worse wetting through lubricant on DLC coated surfaces leads to an earlier formation of slip at the wall between lubricant and surface. This leads to a reduced shear rate in the lubricating gap in the fluid friction regime compared to uncoated surfaces. Kalin et al. investigated the connection between the wetting behaviour of steel and DLC coated surfaces (a-C:H) and the resulting frictional behaviour for dominating fluid friction [9, 10]. Measurements of surface energy and contact angle confirmed the worse wetting behaviour of DLC coatings compared to steel surfaces. A direct connection between the wetting and friction behaviour is established from the experimental results in an oscillating ball on disc tribometer. The evaluation of Stribeck curves confirmed the expectation that higher wall slip for worse wetting does not only lead to a reduction in friction, but also to a decrease in the hydrodynamic load capacity.

Björling et al. [11] identified the thermal conductivity of solid bodies in the near-surface area as further determining factor for the frictional behaviour of lubricated contacts for dominating fluid friction. A rotating ball on disc tribometer is used for experiments in the fluid friction regime for different temperatures, velocities and values of slip. Therefore, uncoated and DLC coated (a-C:Cr) balls and discs were used. Comparable to the results in [5, 8–10], a reduction in friction for DLC coated samples was shown. In addition to the experimental study, a one-dimensional calculation model for the determination of the contact temperature was implemented. The model enables the depth-dependent consideration of different thermal conductivities of the DLC coating and steel. Due to the significant smaller thermal conductivity of DLC, for the DLC coated samples, consistently higher contact temperatures are calculated. According to the authors, the resulting reduction in the lubricant viscosity is responsible for the experimentally observed reduction in friction for DLC coated test samples. Habchi [12] studied the influence of coatings on the behaviour of thermal elastohydrodynamic point contacts with a three-dimensional finite element program. He not only assigned the coatings different thermal physical properties, but also different elastic properties. According to Habchi, coatings with a lower heat conductivity lead to an increase in temperature and therefore to a decrease in fluid friction. The effect was increased with higher coating thickness and slip values. His calculations also showed that coatings with a better thermal conductivity lead to a decrease in temperature and therefore to an increase in friction. The calculation clarifies that the influence of the elastic properties of the coating gains in importance for higher coating thicknesses. In a recent publication of Björling and Habchi [13], the measurements on the ball on disc tribometer used in [11] are compared to the calculation results from the calculation program presented in [12]. As expected, the experiments, as well as the calculation, showed lower friction values using DLC coated test samples. However, bigger differences between the dependence of measured and calculated slip were determined. In the experiments, a consistently lower level of friction was measured, which was mainly related by the authors to the unconsidered gradual warming of the entire system.

The presented studies about the influence of DLC coatings on the frictional behaviour in the fluid friction regime dealt with ideal point contacts. For practical use, elliptic contacts (e.g. rolling bearings) or line contacts (e.g. gear drives) from studies documented below are much more relevant. Hinterstoißer [14] studied the average friction coefficient in the mesh of DLC coated (a-C:H und a-C:H:Me) spur gears in experiments. In particular, with increasing pitch line velocity and therefore increasing fluid friction ratio, he found a clear decrease in friction using a pair with coated surfaces compared to an uncoated reference variant. He ascribes the reduction in friction to an enhanced contact temperature, due to the thermal insulating effect of DLC coatings. Additionally, Hinterstoißer studied the combination of uncoated and DLC coated test gears. In the fluid friction regime, he found a reduction in friction half the size compared to the combination of two coated surfaces. Hinterstoißer concluded, after setting a balance of forces for the integrally considered lubricated contact, that the wall slip on the coated surface cannot be the reason for the reduction in friction. For wall slip on the coated surface, thus, a reduction in friction for the combination of two coated surfaces should be the same as for the combination of coated-uncoated surfaces. The enhanced loss saving for the combination of two coated surfaces can be attributed to the doubled thermal insulating effect of both surfaces. Mayer [15] studied the frictional behaviour of DLC coated (a-C:H:Me und nc-ZrC) surfaces in the lubricated line contact on a twin-disc machine. Based on estimated thermophysical data, he calculated the friction coefficient in fully lubricated disc contacts by means of the limiting shear stress approach of Bair and Winer [16]. The results of experiment and calculation in [15] showed two effects which had a positive influence on the frictional behaviour. First the thermal insulation effect of the surface coatings leads to friction reducing higher lubrication temperatures in the contact. Furthermore, the heat input in the solid body is reduced as a consequence of the thermal insulation, resulting in lower bulk temperatures. The bulk temperatures determine significantly the film thickness in the parallel gap of the rolling contact so that higher film thicknesses result for lower bulk temperatures. For a given sliding speed, an increase in the film thickness leads to a reduction in shear rate. As long as the lubricant's maximum transferable shear stress is not exceeded, a reduction in shear rate also results in a reduction in the friction force.

The objective of this paper is to investigate the often occurring case of line contacts. The experimental and theoretical investigation of the influence of DLC coatings on the friction behaviour of a rolling line contact in the fluid friction regime, including slightly mixed friction, is presented. The experiments are performed using a twin-disc machine with uncoated and DLC coated (a-C:H) samples. In dependence of the slip, the resulting friction forces and the bulk temperature for different parameters in the contact are measured for the driving disc. For the theoretical investigation of the twin-disc contact, a three-dimensional thermal elastohydrodynamic contact is implemented. The TEHL calculation model is based on the combined solution of the Reynolds equation, the elastic half-space and the energy equation of the lubricating gap, as well as Fourier's heat conductivity equation for the determination of the temperature distribution in the solid bodies. Furthermore, the calculation model includes algorithms that consider mass-conserving cavitation and the non-Newtonian flow of the fluid. The heat conduction in solid bodies is calculated, considering the real coating structure. The presented experiments and calculations give a good idea of the effectiveness of DLC coatings for the reduction in friction in thermal elastohydrodynamic line contacts for predominant fluid friction.

2 DLC Coatings

The DLC coating was deposited on 16MnCr5 steel substrates, see Fig. 1. Prior to deposition, the substrates were degreased using a fully automated ultrasonic cleaning line. The coating process started with an ion cleaning followed

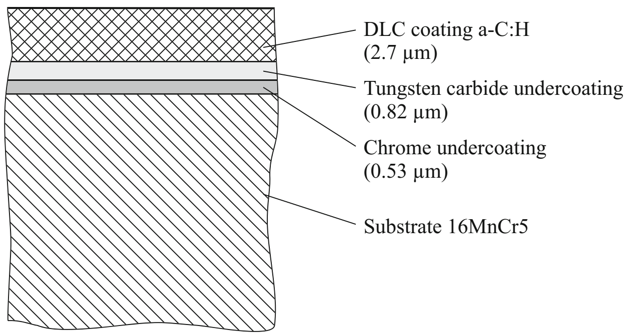


Fig. 1 Coating structure of the DLC coated discs

by deposition of chrome and tungsten carbide in an unbalanced physical vapour deposition (PVD) magnetron sputtering process. The functional a-C:H coating is deposited in a plasma-enhanced chemical vapour deposition (PECVD) step. Acetylene was used at the meantime of deposition since the hardness is higher using acetylene as process gas instead of methane [17]. In the last step, a layer was deposited using methane as process gas. Thereby the tribological run-in behaviour of the coating was optimised. The coating parameters are summarised in Table 1.

3 Experimental Method

3.1 Twin-Disc Machine

The experimental set-up of the twin-disc machine, which is used for friction measurement, is illustrated in Fig. 2. Both test discs were separately driven by two three-phase motors and by infinitely variable friction gears. The test shafts with rolling bearings, on which the test discs were mounted by press fit, were driven by a drive shaft. The rotational speed range varied from 0 to 3500 min^{-1}. The axis-centre distance was set to 80 mm. Thus, for the discs, circumferential speeds of up to 14 m/s were possible.

The upper shaft was mounted in a skid, which was connected to the frame by the use of two steel sheets. A load cell supported the skid laterally, so the frictional force as reaction force between the two discs for slip $S > 0$ %

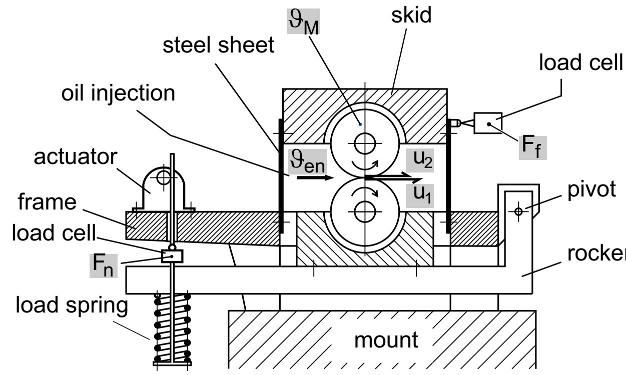


Fig. 2 Twin-disc machine

was measured almost without displacement. This measurement principle allowed a direct measurement of the traction part of the friction. A spring pressed the lower disc, which was mounted in a rocker, against the upper disc. The spring was preloaded by an electrically controllable spindle drive. The spring force was measured by a load cell. The test rig offered all necessary possible adjustments to make sure that the axis of the discs was parallel in two planes (axis cant and axis inclination). Furthermore, the vertical offset of the disc axis was compensated to prevent that the normal force generated a tangential force, which was measured as apparent friction force. To ensure that the load was evenly distributed over the surface, the load distribution was determined using a contact cast. If necessary, the load distribution was corrected by adjustment of the position of the axis. The rotational speed of the test shaft was measured by optical fork sensors and impulse counting. The bulk temperature of the upper disc was measured by a Pt-100 temperature sensor 4 mm below the disc surface during the friction measurement. The data for each single measurement were documented at quasi-steady bulk temperature. The bulk temperature was considered to be quasi-steady if the change was below 0.5 °C/min. The lubricant was injected directly into the disc contact. For the oil supply, a separate oil unit was used to control the oil inlet temperature with an accuracy of ±1 °C. A filter with

Table 1 Coating parameters

	Substrate 16MnCr5	Chrome undercoating	Tungsten carbide undercoating	DLC coating a-C:H
Coating thickness	–	0.53 μm	0.82 μm	2.7 μm
Young's modulus	206,000 N/mm^2	–	–	–
Poisson's ratio	0.3	–	–	–
Density	7760 kg/m^3	7190 kg/m^3	15,800 kg/m^3	1860 kg/m^3
Heat conductivity	44 W/(m K)	90 W/(m K)	58 W/(m K)	0.566 W/(m K)
Specific heat capacity	431 J/(kg K)	447 J/(kg K)	283 J/(kg K)	700 J/(kg K)

nominal fineness of 10 µm was used in the lubricating circuit.

3.2 Test Discs

Both discs are of cylindrical shape and form a line contact (Fig. 3). The base material of the discs was 16MnCr5, which was case-carburised. This ensured that at high pressures no plastic deformations occurred and all experiments took place under the same geometrical conditions. The width of the discs was 5 mm. For constant conditions of the surface topography, the discs were circumferentially ground to an arithmetic mean roughness of $Ra \approx 0.10$–0.15 µm and polished to a mean roughness of $Ra \approx 0.04$ µm.

4 Calculation Model

The calculation model presented in the following sections is part of the calculation software Tribo-X, which is a proprietary development and simulates various tribological contacts [18–20].

4.1 Elastohydrodynamics

Figure 4 presents a schematic of the pressure distribution and the lubricating gap properties of a twin-disc contact in the direction of movement (x direction). The discs are modelled according to the test rig configuration as ideal cylinder without a correction in the direction of width. A Cartesian coordinate system is introduced, and the origin is located at the contact point. When incorporating the elastic deformations of the solid bodies and a constant of integration h_0, the film thickness characteristic in the x and y direction can be calculated with the following equation:

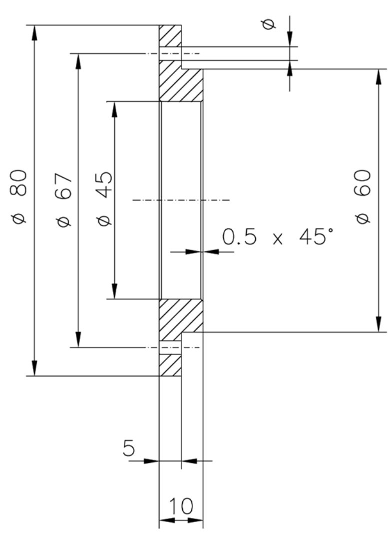


Fig. 3 Test disc

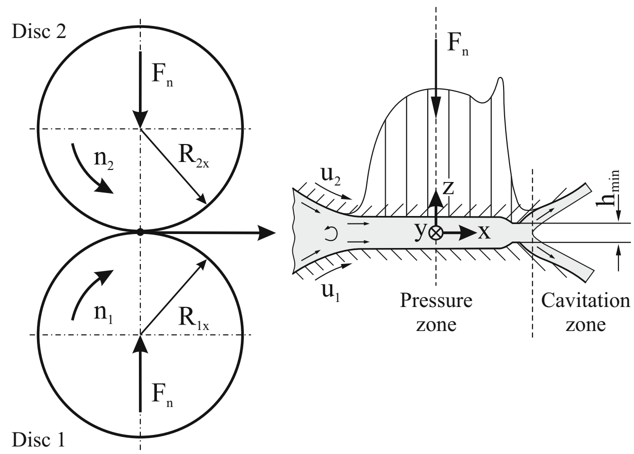


Fig. 4 EHL contact

$$h(x,y) = h_0 + R_{1x}\left(1-\sqrt{1-\left(\frac{x}{R_{1x}}\right)^2}\right) + R_{2x}\left(1-\sqrt{1-\left(\frac{x}{R_{2x}}\right)^2}\right) + \frac{1}{\pi E_{\mathrm{red}}}\iint_{(\Omega)} \frac{p_{\mathrm{h}}(x',y')\mathrm{d}x'\mathrm{d}y'}{\sqrt{(x-x')^2+(y-y')^2}} \quad (1)$$

The calculation of the elastic deformation is based on the theory of the elastic half-space [20]. In consideration of the reduced stiffness at the edges of the solid bodies in these areas, a transformation in the elastic quarter-space is performed [21].

The basis of the elastohydrodynamic calculation model is established by the following generalised Reynolds equation for compressible and laminar flow with an isotropic, non-Newtonian fluid for time-varying film thickness and variable density and dynamic viscosity in the x, y and z direction [20]:

$$\frac{\partial}{\partial x}\left(\Phi^{\mathrm{p}}_{xx}\, G_{1x}\,\frac{\partial p_{\mathrm{h}}}{\partial x}\right) + \frac{\partial}{\partial y}\left(\Phi^{\mathrm{p}}_{yy}\, G_{1y}\,\frac{\partial p_{\mathrm{h}}}{\partial y}\right) = \frac{\partial}{\partial x}\left(\theta\, G_{2x}\,\frac{(u_2-u_1)}{F_0} + \theta\, G_3\, u_1 + \theta\, G_3\, \Phi^{\mathrm{s}}_{xx}\,\frac{(u_2-u_1)}{2\,h}\right) + \frac{\partial}{\partial t}(\theta\, G_3) \quad (2)$$

with $p_h > p_{\mathrm{cav}}$ and $\theta = 1$ in the pressure zone and $p_{\mathrm{h}} = p_{\mathrm{cav}}$ and $\theta < 1$ in the cavitation zone

$$G_{1x} = G_{1y} = \int_0^h \rho_{\mathrm{liq}}(z)\left[\int_0^z \frac{z'}{\eta^*_{\mathrm{liq}}(z')}\,\mathrm{d}z' - \frac{F_1}{F_0}\int_0^z \frac{1}{\eta^*_{\mathrm{liq}}(z')}\,\mathrm{d}z'\right]\mathrm{d}z$$

$$G_{2x} = \int\limits_0^h \rho_{\text{liq}}(z) \left[\int\limits_0^z \frac{1}{\eta^*_{\text{liq}}(z')} \, dz' \right] \quad G_3 = \int\limits_0^h \rho_{\text{liq}}(z) \, dz$$

$$F_0 = \int\limits_0^h \frac{1}{\eta^*_{\text{liq}}(z)} \, dz \quad F_1 = \int\limits_0^h \frac{z}{\eta^*_{\text{liq}}(z)} \, dz$$

The pressure and temperature dependence is calculated according to [21]. The known Vogel equation is used for the acquisition of the temperature dependence of the viscosity [19]. Further the pressure dependence of the viscosity is taken into account by the Roelands equation [23]:

$$\eta_{\text{liq}}(p_h) = \eta(\vartheta) \exp\left[\ln(\eta(\vartheta) + 9.67) \left(-1 + \left(1 + \frac{p_h}{p_0} \right)^z \right) \right] \tag{3}$$

with $p_0 = 1.96 \cdot 10^5$ Pa and $z = \frac{\alpha_p p_0}{\ln(\eta(\vartheta)+9.67)}$

Non-Newtonian flow is expressed as a nonlinear correlation between the hydrodynamic shear stress τ_{fh} and the shear rate in the lubricating gap. Following [20, 24] this action is described by replacing the fluid's dynamic viscosity η_{liq} with a variable, effective viscosity η^*_{liq} in the z direction, which locally retains the formulation of Newton's law of shear stress:

$$\dot{\gamma}_{zx}(z) = \frac{\partial u}{\partial z} = \frac{\tau_{fhzx}(z)}{\eta^*_{\text{liq}}(z)} \quad \text{and} \quad \dot{\gamma}_{zy}(z) = \frac{\partial v}{\partial z} = \frac{\tau_{fhzy}(z)}{\eta^*_{\text{liq}}(z)} \tag{4}$$

Eyring's shear thinning fluid model [25] is used to represent the non-Newtonian flow:

$$\dot{\gamma} = \frac{\tau_c}{\eta_{\text{liq}}} \sinh\left(\frac{\tau_e}{\tau_c} \right) \quad \text{with} \quad \tau_e = \sqrt{\tau^2_{fhzx} + \tau^2_{fhzy}} \tag{5}$$

The validity of the Eyring model decreases with increasing pressures and shear stresses, since no shear stress limit is considered. However, the pressures in the studied contact (see Sect. 5) are not as high, so that the application of Eyring model is still possible. The shear rates in x and y direction of the contact, as introduced in Eq. (4), can be formulated following Eq. (5):

$$\dot{\gamma}_{zx}(z) = \frac{\partial u}{\partial z} = \frac{\tau_{fhzx}(z)}{\eta_{\text{liq}}(z)} \frac{\tau_c}{\tau_e(z)} \sinh\left(\frac{\tau_e(z)}{\tau_c} \right) \tag{6}$$

$$\dot{\gamma}_{zy}(z) = \frac{\partial v}{\partial z} = \frac{\tau_{fhzy}(z)}{\eta_{\text{liq}}(z)} \frac{\tau_c}{\tau_e(z)} \sinh\left(\frac{\tau_e(z)}{\tau_c} \right) \tag{7}$$

Assuming a linear development of the hydrodynamic shear stresses in the z direction of the lubricating gap

$$\tau_{fhzx}(z) = \tau_{fhzx,1} + z \frac{\partial p_h}{\partial x} \quad \text{and} \quad \tau_{fhzy}(z) = \tau_{fhzy,1} + z \frac{\partial p_h}{\partial y} \tag{8}$$

Equations (6) and (7) are integrated as a function of the film thickness h. The solution of the resulting equation system, which is based on the unknown shear stresses at the surface 1 $\tau_{fhzx,1}$ and $\tau_{fhzy,1}$, is used to calculate the curves of the shear stresses according to Eq. (8) and to determine the curve of the effective dynamic viscosity from the Eqs. (4) and (5):

$$\eta^*_{\text{liq}}(z) = \eta_{\text{liq}}(z) \frac{\tau_e(z)}{\tau_c} \sinh\left(\frac{\tau_e(z)}{\tau_c} \right)^{-1} \tag{9}$$

Readers are referred to Liu et al. [26] for further details on the implementation of this model. When the masses are balanced, the validity of the Reynolds Eq. (2) extends to the entire calculation domain of the lubricating gap and, thus, includes regions of pressure and cavitation. To describe the flow characteristics in the cavitation region, an algorithm based on the JFO theory is applied to the calculation model [27]. Therefore, the Reynolds equation is expanded by a gap fill factor θ, which expresses the ratio of the lubricant volume to the entire volume in the cavitation region as $0 < \theta < 1$. The introduction of the gap fill factor in the presented form transforms the flow in the cavitation region into a homogeneous, two-phase flow. Thus, the Reynolds equation is formulated to be a complementary problem for the unknown variables of the hydrodynamic pressure p_h and the gap fill factor θ. The hydrodynamic pressure distribution must be in equilibrium with the external load F_n, so that the force equilibrium is calculated by the following equation:

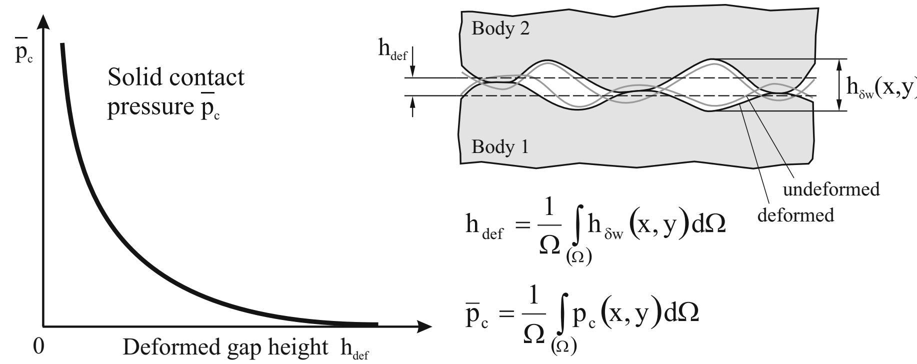
$$F_n = \int\limits_{(\Omega)} p_h(x, y) \, d\Omega \tag{10}$$

4.2 Mixed Friction and Microhydrodynamics

Mixed friction refers to the operating state of simultaneously occurring boundary and fluid friction as well as solid contact pressures and hydrodynamic pressures at a lubricated contact with rough surfaces. The model used for simulations is based on the calculation of local pressure distributions of solid contacts for representative surface sections of the contact areas by means of a discrete elastic plastic solid contact model and the transformation in the integral solid contact pressure curves for the direct application in the thermal elastohydrodynamic calculation model, such as shown in Fig. 5. Readers are referred to [18, 20] for a more detailed description of the calculation model.

A pressure distribution occurs in the mixed friction area of the lubricating gap, which results from the combined

Fig. 5 Integral solid contact pressure (schematic)



distributions of the hydrodynamic pressures and the integral solid contact pressures:

$$p(x,y) = p_{\mathrm{h}}(x,y) + \bar{p}_{\mathrm{c}}(x,y) \tag{11}$$

The resulting shear stress distribution in the moving direction of the contact (x direction) is the result of the hydrodynamic shear stress and the shear stress of the boundary friction:

$$\begin{aligned} &\tau_{fx}(x,y) = \tau_{fhzx}(x,y) + \bar{\tau}_{\mathrm{fc}}(x,y) \quad \text{with} \\ &\bar{\tau}_{\mathrm{fc}}(x,y) = f_{\mathrm{c}} \cdot \bar{p}_{\mathrm{c}}(x,y) \end{aligned} \tag{12}$$

The calculation of the shear stress of the boundary friction is based on the definition of the constant boundary coefficient of friction f_{c}, which is derived from experience. For mixed friction the equilibrium of forces (10) is expanded to:

$$F_{\mathrm{n}} = \int_{(\Omega)} p(x,y)\,\mathrm{d}\Omega = \int_{(\Omega)} \left(p_{\mathrm{h}}(x,y) + \bar{p}_{\mathrm{c}}(x,y)\right)\mathrm{d}\Omega \tag{13}$$

For film thicknesses in the range of the roughness of the contact surfaces, the influences of microhydrodynamic effects on the hydrodynamic pressure development cannot be neglected. Building upon the studies of Patir and Cheng [28], the Reynolds Eq. (2) is expanded by two pressure flow factors Φ^{p}_{xx} and Φ^{p}_{yy}, and a shear flow factor Φ^{s}_{xx} to consider the microhydrodynamic effects. The flow factors are calculated numerically based on the real measured surface topographies. Solid contacts and periodical boundary conditions are considered. Readers are referred to [18, 20] for a more detailed description of the calculation model.

4.3 Temperature

The three-dimensional temperature distribution in the lubricating gap is calculated with aid of the transient energy equation for the fluid, which is used to establish the equilibrium of heat sources and sinks (14). Fluid friction and compression processes constitute heat sources. Convection, heat conduction and expansion processes are heat sinks.

$$\begin{aligned} &\underbrace{\beta_{\mathrm{th,mix}}\,\vartheta\left(\frac{\partial p_{\mathrm{h}}}{\partial t} + u\frac{\partial p_{\mathrm{h}}}{\partial x} + v\frac{\partial p_{\mathrm{h}}}{\partial y}\right)}_{\text{Compression/Expansion}} + \underbrace{\tau_{fhzx}\frac{\partial u}{\partial z} + \tau_{fhzy}\frac{\partial v}{\partial z}}_{\text{Fluid friction}} = \\ &\underbrace{c_{\mathrm{p,mix}}\,\rho_{\mathrm{mix}}\left(\frac{\partial \vartheta}{\partial t} + u\frac{\partial \vartheta}{\partial x} + v\frac{\partial \vartheta}{\partial y} + w\frac{\partial \vartheta}{\partial z}\right)}_{\text{Convection}} \\ &\underbrace{-\lambda_{\mathrm{mix}}\left(\frac{\partial^2 \vartheta}{\partial x^2} + \frac{\partial^2 \vartheta}{\partial y^2} + \frac{\partial^2 \vartheta}{\partial z^2}\right)}_{\text{Heat conduction}} \end{aligned} \tag{14}$$

Polynomial functions based on discrete values are implemented to ascertain the correlations of the thermophysical variables of the lubricant with the pressure and temperature. In the presented form, the energy equation applies to the entire calculation domain of the lubricating gap. For the description of the thermophysical variables, a linear mixing approach for the combination of fluid and gas-vapour phases which is based on the assumption that a homogeneous, two-phase flow in the cavitation region exists, is applied [18]. The calculation of the temperatures at the surfaces and inside the bodies is based on the transient Fourier heat equation:

$$\underbrace{c_{v,\mathrm{solid}}\,\rho_{\mathrm{solid}}\left(\frac{\partial \vartheta}{\partial t}\right)}_{\text{Transient term}} = \underbrace{\lambda_{\mathrm{solid}}\left(\frac{\partial^2 \vartheta}{\partial x^2} + \frac{\partial^2 \vartheta}{\partial y^2} + \frac{\partial^2 \vartheta}{\partial z^2}\right)}_{\text{Heat conduction}} \tag{15}$$

Solid bodies and lubricant are interrelated as a function of the heat flux densities at the surfaces:

$$\dot{q}_{\mathrm{liq(mix)}} + k\,|u_{\mathrm{rel}}\,\tau_{\mathrm{fc}}| = \dot{q}_{\mathrm{solid}} \quad \text{with} \quad \dot{q} = \lambda\frac{\partial \vartheta}{\partial z} \tag{16}$$

The transition condition between fluid and solid bodies is supplemented by an additional heat flux density as the product of the coefficient of thermal partition k, the relative velocity $|u_{\mathrm{rel}}| = |u_1 - u_2|$ and the shear stress of the boundary friction τ_{fc}, to ascertain the additional heating of the solid bodies that results from the boundary friction during mixed friction. The coefficient of thermal partition describes the relative part of heat flux density in a particular body, thus, $k_1 + k_2 = 1$ ultimately applies to both bodies [18]. As boundary condition, a constant bulk temperature in a defined depth of the solid body is given. The application of the Fourier heat equation for the calculation of the temperature distribution of the solid bodies allows the specification of discrete variable thermophysical material properties. With this the thermophysical behaviour of multi-layered solid bodies can be represented. The accuracy of the discretisation in the model is more flexible for the individual layer which enables the representation of different film thicknesses.

4.4 Numerics

The Reynolds Eq. (2) is solved numerically, in consideration of the mass conservation cavitation model based on an iterative method proposed in [29]. High numerical efficiency is attained by consequently applying multigrid methods. The equations that calculate the elastic deformations of the solid bodies, the rheological properties of the lubricant and the characteristics of mixed friction and microhydrodynamics are solved simultaneously. In parallel, the energy Eq. (14) and the Fourier heat Eq. (15) are solved iteratively, with finite difference discretisation.

5 Results

The twin-disc machine, such as introduced in Sect. 3, was used for the experiments. Furthermore, calculations using the thermal elastohydrodynamic calculation model (see Sect. 4) were performed. The experimental parameters are summarised in Table 2. The objective of the experiments was to investigate the frictional behaviour for different experimental parameters in dependence of the slip between the discs. Therefore, the slip between the discs was increased stepwise from 0 up to 50 %, in which every test point was hold until the quasi thermal steady-state condition was reached. All experiments were performed with constant load and sum velocity. The slip, oil supply temperature and the coating condition of the discs were varied. The presented experiments are summarised in Table 3, and the significant oil data are summarised in Table 4. The structure of the DLC coating and the material data of the discs can be found in Sect. 2.

The operated surfaces of the discs were measured three dimensionally with a lateral point distance of 0.2 µm in order to apply the models that were presented in Sect. 4.2 to mixed friction and microhydrodynamics. Figure 6 presents the surface sections of the uncoated and DLC coated discs. Both discs had plateau like surface topographies with scoring structure in circumferential direction. In order to get dominating fluid friction, the asperities were kept small. The measurements delivered a critical film thickness of $h_{cr} = 0.19$ µm for the experiments with the uncoated discs (experiments 1 and 2 after Table 3) and of $h_{cr} = 0.25$ µm for the experiments with the DLC coated discs (experiments 3 and 4 after Table 3).

The characteristics of the integral solid contact pressures and flow factors were calculated to be a function of the deformed gap height and are presented in Fig. 7 for the combinations of uncoated and DLC coated discs. A plastic flow pressure of $p_{c,lim} = 7250$ N/mm^2 for the uncoated discs and of $p_{c,lim} = 8100$ N/mm^2 for the DLC coated discs was ascertained from measurements of Martens hardness with a maximum test force of 1 N, to be the condition for plastic flow for the asperities. For distances lower than the critical gap height h_{cr}, first solid contact

Table 2 Experimental parameter of the twin-disc machine

Base material disc 1	16MnCr5
Base material disc 2	16MnCr5
Coating	Uncoated and DLC coated
Diameter of the disc	$d = 80$ mm
Width of the disc	$b = 5$ mm
Correction in the direction of width	None
Load	$F = 3921$ N
Maximum Hertzian stress	$p_0 = 1189$ N/mm^2
Sum velocity	$u_\Sigma = u_1 + u_2 = 8$ m/s
Slip	$S = (u_1 - u_2)/u_1 = 0 \ldots 50$ %
Lubricant	FVA 3 (ISO VG 100)
Oil supply temperature	$\partial_{en} = 40$ and 100 °C

Table 3 Summary of the experiments

Experiment	Disc 1	Disc 2	Oil supply temperature (°C)
1	Uncoated	Uncoated	40
2	Uncoated	Uncoated	100
3	DLC coated	DLC coated	40
4	DLC coated	DLC coated	100

Table 4 Oil data FVA3

Density (15 °C)	$\rho_{15} = 880$ kg/m^3
Dynamic viscosity (40 °C)	$\eta_{40} = 82.09$ mPas
Dynamic viscosity (100 °C)	$\eta_{100} = 8.21$ mPas
Specific heat capacity (40 °C)	$c_{p40} = 2258.8$ J/(kg K)
Heat conductivity (40 °C)	$\lambda_{40} = 0.145$ W/(m K)
Eyring shear stress	$\tau_c = 6$ N/mm^2

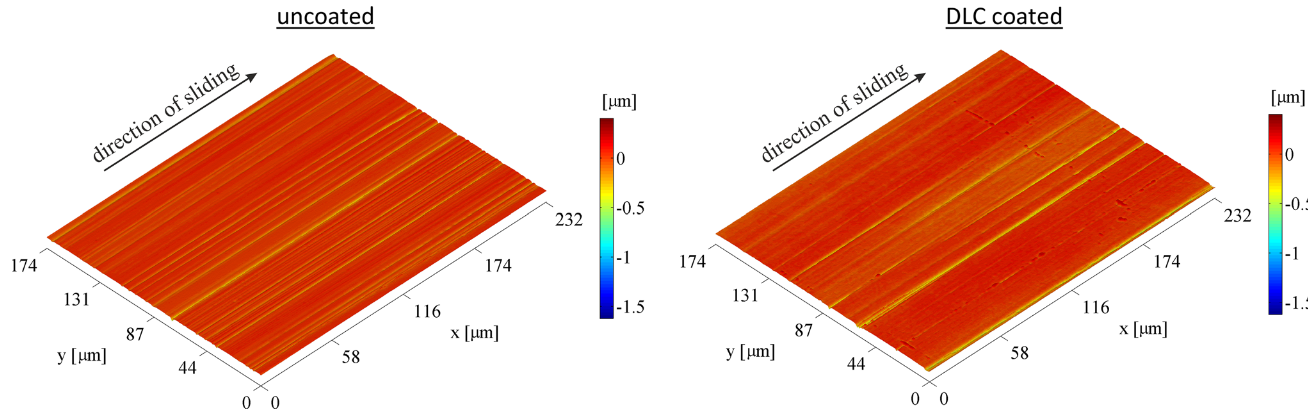


Fig. 6 Surface topographies of uncoated and DLC coated discs (examples)

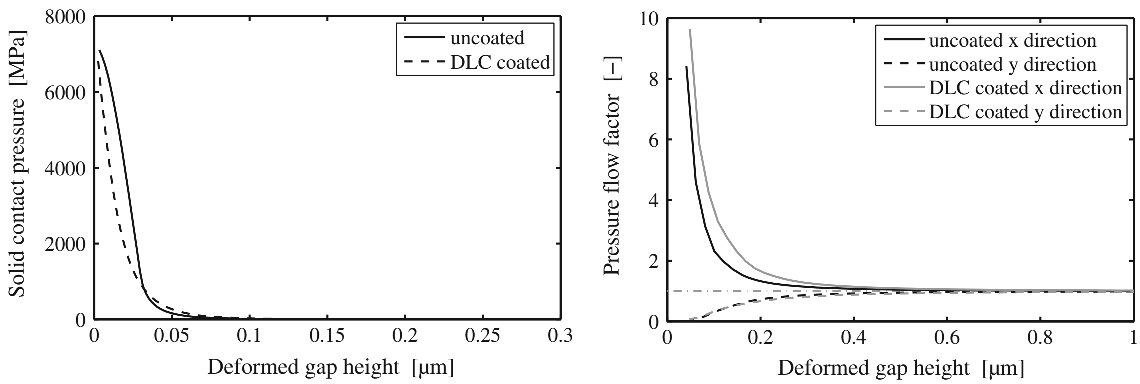


Fig. 7 Integral solid contact pressure and flow factors

pressures can occur. Beyond that, a nonlinear increase with decreasing gap height to the plastic flow pressure occurred. The solid contact pressures were in the same range for both combinations. The influence of the scoring structure of the surface topographies as an impediment to the flow in the x direction and an improvement of the flow in y direction was discernible in the curves of the pressure flow factors. The coefficient of boundary friction, for the determination of the shear stress of the boundary friction (Eq. 12), was set to $f_c = 0.05$ for both combinations of discs. The choice of the coefficients based on experience. A differentiated determination of the coefficients of boundary friction was not performed, since the influence of the DLC coating on the frictional behaviour at dominating fluid friction was in the foreground of the investigations. The influence of the DLC coating on the boundary friction was thus deliberately neglected.

The calculation of the twin-disc contact was performed at a resolution of $145 \times 201 \times 270$ points (x-y-z direction). In z direction, the number of points is uniformly dispersed on the lubricating gap and both solid bodies. Due to the different near-surface heat conductivities, the uncoated discs were discretised to a depth of 0.5 mm and the DLC coated discs to 0.05 mm, with variable and higher point distances in the depth direction. For the DLC coated discs, the point distance at the beginning of a new layer (see Fig. 1) was reduced, in order to get the expected higher temperature gradient in the area of the film transition. The required boundary condition for the temperature of the discs at the deepest point of the discretisation was set according to the measured bulk temperature of disc 1. As boundary condition for the temperature of the inflowing oil in the lubricating gap, the measured bulk temperature was taken as well. This assumption is reasonable if it can be assumed that even before the contact area temperature compensation between lubricant and discs occurs. The different material properties of the base material compared to the properties of the DLC coating were not included for the calculation of the elastic deformations. This simplification is permitted for thin coatings [12]. The calculation domain was set to 2 mm in the x direction and to 5 mm in the y direction, which corresponds to the width of the discs. The calculations were performed in the slip range from 0 % up to 50 % at 101 uniformly dispersed calculation points.

Figure 8 presents the curves of the friction force and the maximum temperature plotted against the slip for the experimental and calculation results of the twin-disc

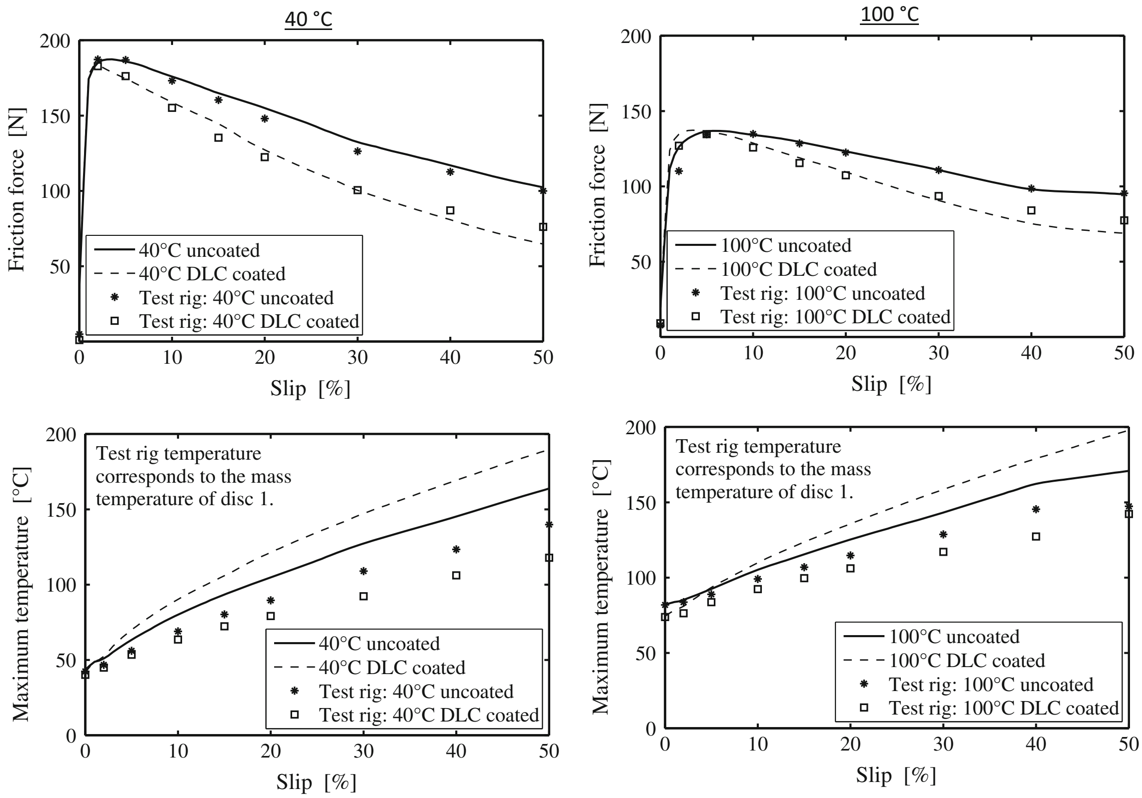


Fig. 8 Results for friction force and maximum temperature for 40 and 100 °C

contact. The friction forces refer to disc 1. The comparison between measured and calculated friction force for both oil supply temperatures showed a good agreement. The maximum temperature of the test rig is the bulk temperature of disc 1 during the experiment. The calculated maximum temperature shows the highest locally determined temperature in the lubricating gap at the corresponding time. The measured, as well as the calculated friction forces, increased with increasing slip, since the increase in the slip resulted in an increase in the frictional shear stress in the contact. After reaching a maximum, the friction forces decreased with increasing slip. This behaviour is the result of the continuously increasing temperature in the contact, caused by friction and the connected decrease in the effective lubricant viscosity. Until reaching the maximum of friction, the friction forces of the uncoated and DLC coated discs were at the same level. Afterwards, the DLC coated discs had smaller friction forces, whereby the difference with increasing slip increases continuously. The reason for this behaviour is the different temperature characteristic in the lubricating gap for the uncoated and DLC coated discs. The lower heat conductivity of DLC compared to steel (see Table 1) caused a smaller heat removal from the lubricating gap in the discs. The resulting higher temperature led, due to the decrease in the effective viscosity of the lubricant, to the investigated friction difference between uncoated and DLC coated discs, both, in experiment and calculation. This behaviour became visible by means of the measured bulk temperatures from the discs. These were lower for the DLC coated discs, despite the higher temperature in the lubricating gap. The increase in the nominal oil supply temperature from 40 to 100 °C resulted in a decrease in the measured and calculated friction levels and in an increase in the measured and calculated bulk temperature. Figure 9 presents the calculated curves of the minimum film thickness and the solid contact load ratio plotted against the slip. The minimum film thickness decreased with increasing slip due to the temperature increase in the lubricating gap. The characteristics of the curves of the solid contact load ratio showed that the experiments were partially performed under light mixed friction conditions. However, the solid contact load ratios for both analysed oil temperatures were quite small so that the objective to investigate the system for dominating fluid friction was not offended.

Figure 10 presents the local distributions of the film thickness and of the solid contact pressure for the experiments at 100 °C oil supply temperature with 50 % slip. At this operating point the highest solid contact load ratios occurred. The film thickness was characterised by strong restrictions at the edges of the lubricating gap. The solid contact pressures also reached their maximum at these

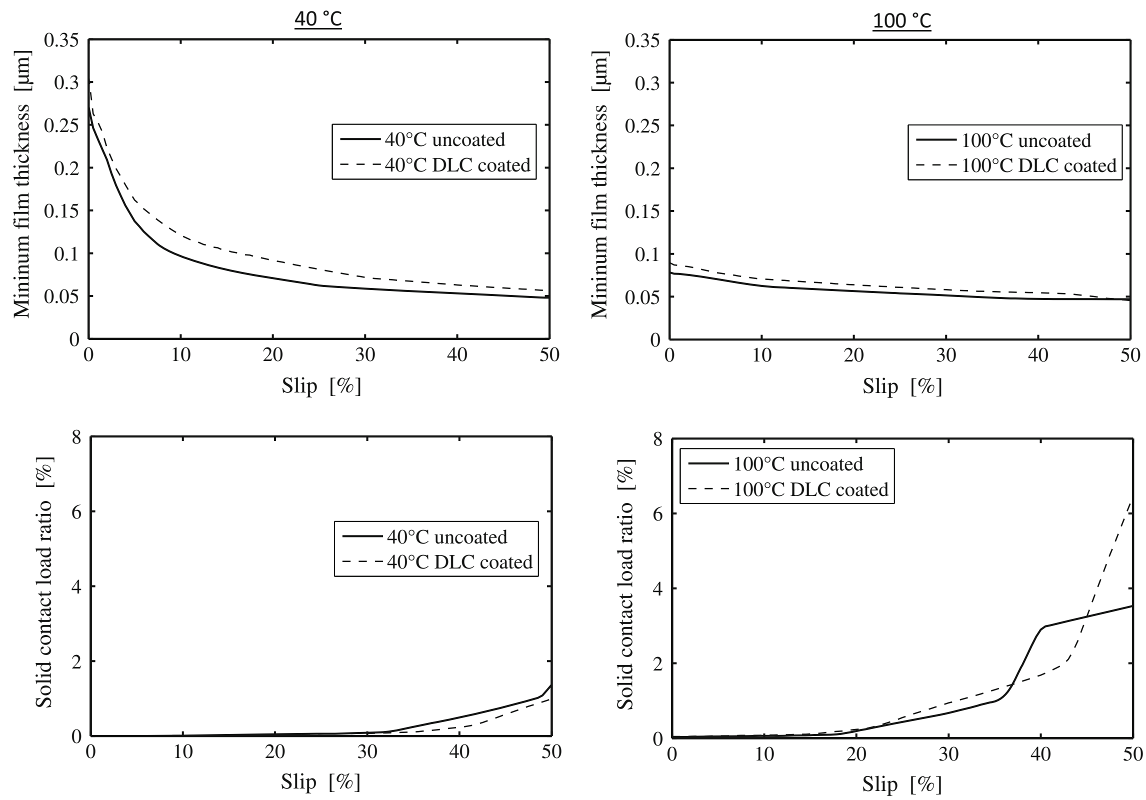


Fig. 9 Results for minimum film thickness and solid contact load ratio for 40 and 100 °C

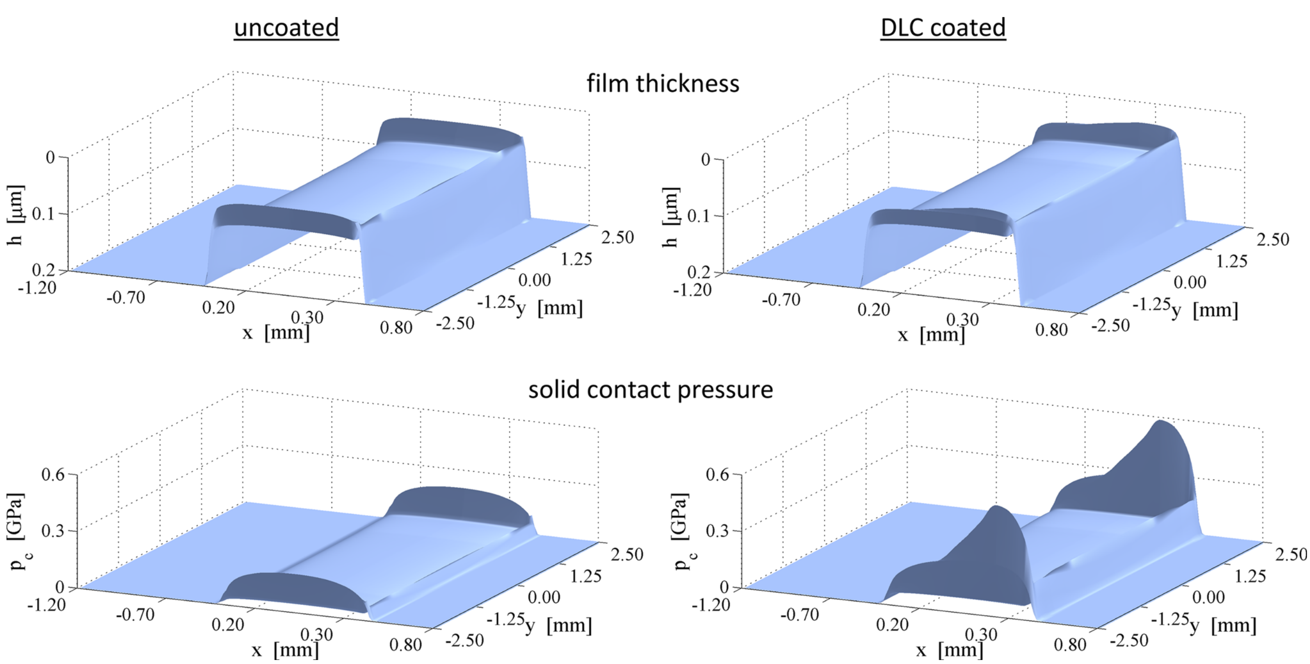


Fig. 10 Local results (3D) for film thickness and solid contact pressure for 100 °C at 50 % slip

edges. Since the DLC coated discs had a higher surface roughness compared to the uncoated discs, the highest solid contact pressures were reached for DLC coated discs.

Despite the higher maximum temperatures in the lubricating gap, the characteristics of minimum film thickness for the combination of DLC coated discs (Fig. 9) were

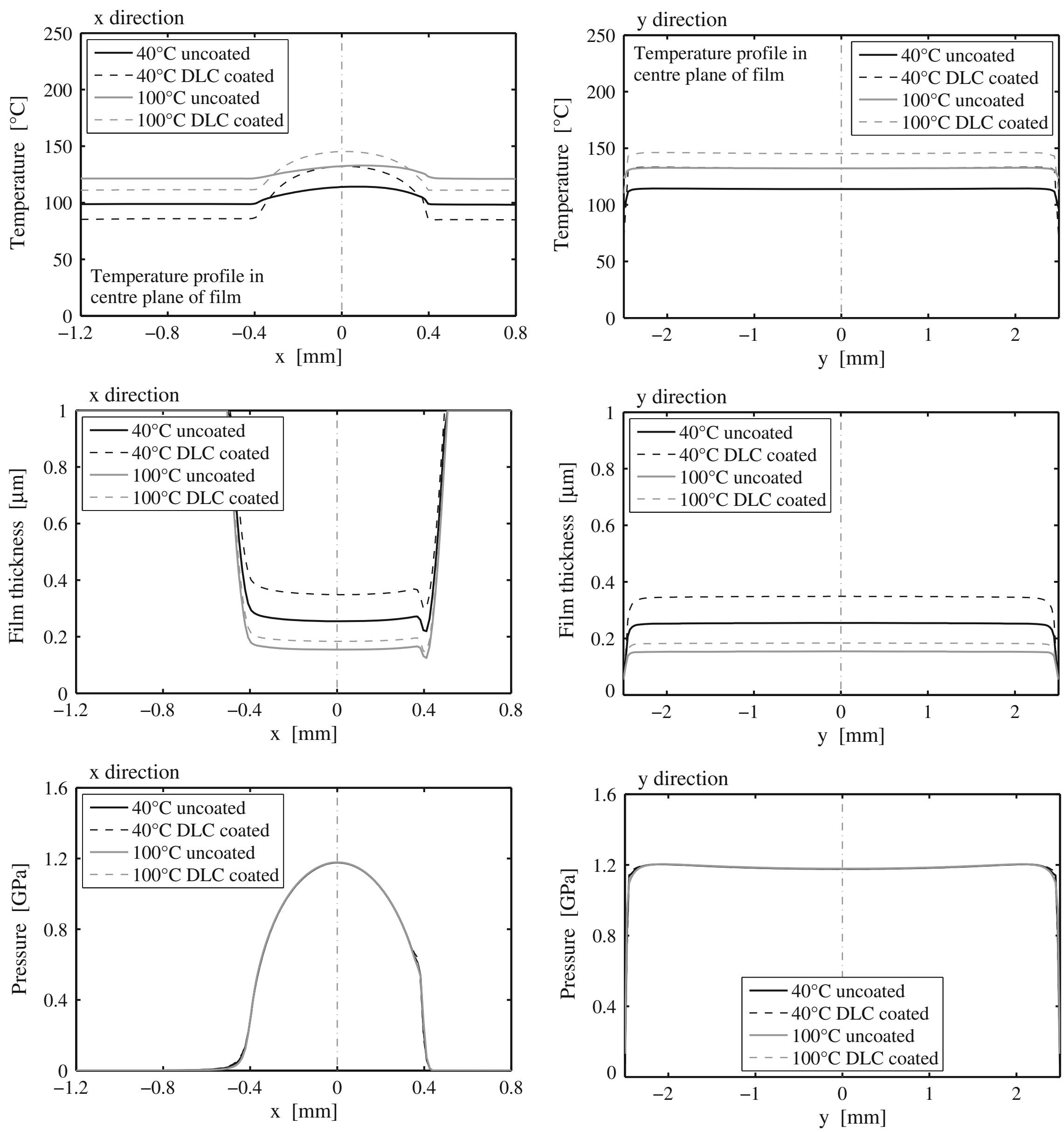


Fig. 11 Local results (2D) for temperature, film thickness and pressure for 40 and 100 °C at 25 % slip (section in symmetry plane)

higher compared to the uncoated discs. Due to the lower temperatures at the inlet zone of the lubricating gap, compared to the combination of the uncoated discs, for the DLC coated discs higher lubricating gaps were reached. The temperature difference is the consequence of different bulk temperatures, such as shown in Fig. 8. This behaviour is shown in Fig. 11 for a slip of 25 % in the temperature, film thickness and pressure curves. The temperature at the inlet zone of the contact is lower for the combination of DLC coated discs, compared to the uncoated discs. The sequence changes in the middle of the contact, due to the lower heat conductivity of the DLC coating. The viscosity at the inlet zone of the contact is formative for the lubricating gap at the entire contact, resulting in higher film thicknesses for the DLC coated discs. Another reason for the friction reduction by use of DLC coatings is the so resulting smaller shear rates in the contact. Figure 11 presents the independence of the pressure distribution of the coating and the oil supply temperatures.

6 Conclusion

The influence of DLC coatings on the frictional behaviour of lubricated line contacts is investigated, using a twin-disc machine and thermal elastohydrodynamic simulations. Uncoated and DLC coated discs are used for the experiments. The experiments are performed in the fluid friction regime with a variation of slip for two different oil supply temperatures. The results from experiments are in line with the results from the calculations. For fluid friction, a friction reduction in thermal elastohydrodynamic contacts is achieved by use of DLC coatings. The smaller heat conductivity of DLC coatings is identified as the main source for the reduction in friction. The resulting higher temperature level in the lubricating gap results in a reduction in the effective viscosity of the lubricant and therefore in lower frictional shear stress. This model conception is confirmed by measurement of the bulk temperature during the experiments. For the same experimental parameter set, the DLC coated discs have lower bulk temperatures than the uncoated discs. Furthermore, it is shown that lower bulk temperatures of the DLC coated discs resulted in higher film thicknesses in the contact, which in turn results in a friction reduction in the lubricating gap. The possible influence of the wetting behaviour of uncoated and coated surfaces is not considered in the calculation model. The drawn conclusions from the reduction in friction in lubricated contacts due to DLC coatings refer to friction in the fluid friction regime. The boundary friction between the surfaces cannot be ignored for pronounced mixed friction. The contribution of DLC coatings on the reduction in friction then depends on the material parameters of the DLC coating and the interaction with the lubricant and the lubricant's additives.

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References

1. VDI-Society Production und Logistics: Carbon films: basic knowledge, film types and properties. VDI-Guideline, 2840 (2012)
2. Illner, T.: Oszillierendes kippbewegliches Axialgleitlager bei Grenzreibung und Kraftstoffschmierung. PhD thesis. University of Magdeburg (2010)
3. Kalin, M., Vizintin, J.: The tribological performance of DLC-coated gears lubricated with biodegradable oil in various pinion/gear material combinations. Wear **259**, 1270–1280 (2005)
4. Jahanmir, S., Hunsberger, A.Z., Heshmat, H.: Load capacity and durability of H-DLC coated hydrodynamic thrust bearings. J. Tribol. **133**, 031301 (2011)
5. Evans, R.D., Cogdell, J.D., Richter, G.A., Doll, G.L.: Traction of lubricated rolling contacts between thin-film coatings and steel. Tribol. Trans. **52**, 106–113 (2009)
6. Yang, L., Neville, A., Brown, A., Ransom, P., Morina, A.: Friction reduction mechanisms in boundary lubricated W-doped DLC coatings. Tribol. Int. **70**, 26–33 (2014)
7. Kogovsek, J., Remskar, M., Kalin, M.: Lubrication of DLC-coated surfaces with MoS_2 nanotubes in all lubrication regimes: surface roughness and running-in effects. Wear **303**, 361–370 (2013)
8. Grossl, A.: Einfluss von PVD-Beschichtungen auf die Flanken- und Fußtragfähigkeit einsatzgehärteter Stirnräder. PhD thesis. Technical University of Munich (2007)
9. Kalin, M., Velkavrh, I., Vizintin, J.: The Stribeck curve and lubrication design for non-fully wetted surfaces. Wear **267**, 1232–1240 (2009)
10. Kalin, M., Polajnar, M.: The effect of wetting and surface energy on the friction and slip in oil-lubricated contacts. Tribol. Lett. **52**, 185–194 (2013)
11. Björling, M., Isaksson, P., Marklund, P.: The influence of DLC coating on EHL friction coefficient. Tribol. Lett. **47**, 285–294 (2012)
12. Habchi, W.: A numerical model for the solution of thermal elastohydrodynamic lubrication in coated circular contacts. Tribol. Int. **73**, 57–68 (2014)
13. Björling, M., Habchi, W., Bair, S., Larsson, R., Marklund, P.: Friction reduction in elastohydrodynamic contacts by thin-layer thermal insulation. Tribol. Lett. **53**, 477–486 (2014)
14. Hinterstoißer, M.: Zur Optimierung des Wirkungsgrades von Stirnradgetrieben. PhD thesis. Technical University of Munich (2014)
15. Mayer, J.: Einfluss der Oberfläche und des Schmierstoffs auf das Reibungsverhalten im EHD-Kontakt. PhD thesis. Technical University of Munich (2013)
16. Bair, S., Winer, W.O.: A rheological model for elastohydrodynamic contacts based on primary laboratory data. J. Lubr. Technol. **101**, 258–265 (1979)
17. Robertson, J.: Diamond-like amorphous carbon. Mater. Sci. Eng. **37**, 129–282 (2002)
18. Bobach, L., Beilicke, R., Bartel, D., Deters, L.: Thermal elastohydrodynamic simulation of involute spur gears incorporating mixed friction. Tribol. Int. **48**, 191–206 (2012)
19. Bartel, D., Bobach, L., Illner, T., Deters, L.: Simulating transient wear characteristics of journal bearings subjected to mixed friction. J. Eng. Tribol. **226**, 1095–1108 (2012)
20. Bartel, D.: Simulation von Tribosystemen—Grundlagen und Anwendungen, 1st edn. Vieweg+Teubner, Wiesbaden (2010)
21. Guilbault, R.: A fast correction for elastic quarter-space applied to 3D modeling of edge contact problems. J. Tribol. **133**, 031402 (2011)
22. Bode, B.: Modell zur Beschreibung des Fließverhaltens von Flüssigkeiten unter hohem Druck. Tribol. Schmierungstech. **36**, 182–189 (1989)
23. Roelands, C.J.A.: Correlational aspects of the viscosity-temperature-pressure relationship of lubricating oil. PhD thesis. Technical University Delft (1966)
24. Yang, P., Wen, S.: A generalized reynolds equation for non-newtonian thermal elastohydrodynamic lubrication. J. Tribol. **112**, 631–636 (1990)
25. Eyring, H.: Viscosity, plasticity and diffusion as examples of absolute reaction rates. J. Chem. Phys. **4**, 283–291 (1936)
26. Liu, X., Jiang, M., Yang, P., Kaneta, M.: Non-newtonian thermal analyses of point EHL contacts using the Eyring model. J. Tribol. **127**, 70–81 (2005)
27. Elrod, H.G.: A cavitation algorithm. J. Lubr. Technol. **103**, 350–354 (1981)
28. Patir, N., Cheng, H.S.: Application of average flow model to lubrication between rough sliding surfaces. J. Lubr. Technol. **101**, 220–230 (1979)
29. Venner, C.H.: Multilevel solution of the EHL line and point contact problems. PhD thesis. University of Twente (1991)